For reasons similar to these, astronomers might *never* find any kind of definitive black-hole signature in any electromagnetic waves produced from the vicinity of a black hole. Fortunately, the prospects are excellent for a completely different kind of black-hole signature: one carried by gravitational radiation. To this we shall return in Chapter 10.

☆ ☆ ☆

The golden age of theoretical black-hole research (Chapter 7) coincided with the observational search for black holes and the discovery of Cyg X-1 and deciphering of its nature. Thus, one might have expected the youths who dominated the golden age (Penrose, Hawking, Novikov, Carter, Israel, Price, Teukolsky, Press, and others) to play key roles in the black-hole search. Not so, except for Novikov. The talents and knowledge that those youths had developed, and the remarkable discoveries they were making about black-hole spin, pulsation, and hairlessness, were irrelevant to the search and to deciphering Cyg X-1. It might have been different if Cyg X-1 had had an unequivocal black-hole signature. But there was none.

These youths and other theoretical physicists like them are sometimes called *relativists,* because they spend so much time working with the laws of general relativity. The theorists who *did* contribute to the search (Zel'dovich, Paczynski, Sunyaev, Rees, and others) were a very different breed called *astrophysicists.* For the search, these astrophysicists needed to master only a tiny amount of general relativity—just enough to be confident that curved spacetime was quite irrelevant, and that a Newtonian description of gravity would be quite sufficient for modeling an object like Cyg X-1. However, they needed enormous amounts of *other* knowledge, knowledge that is part of the standard tool kit of an astrophysicist. They needed a mastery of extensive astronomical lore about binary star systems, and about the structures and evolutions and spectra of the companion stars of black-hole candidates, and about the reddening of starlight by interstellar dust—a key tool in determining the distance to Cyg X-1. They also needed to be experts on such issues as the flow of hot gas, shock waves formed when streams of hot gas collide, turbulence in the gas, frictional forces in the gas caused by turbulence and by chaotic magnetic fields, violent breaking and reconnection of magnetic field lines, the formation of X-rays